

1. install git.

Purpose : To install Git and verify that Git is available on the local system.

Command : git --version

Expected Output

git version 2.37.1

Description : The git --version command displays the installed Git version. This confirms that Git is successfully installed and ready to use.

```
chvka@karthik MINGW64 ~/git-task
$ git --version
git version 2.37.1.windows.1
```

2. Create a repo in github with README.md and .ignore file.

Purpose : To create a central repository on GitHub and initialize it with basic repository files.

Steps:

Open GitHub.

Click New Repository.

Enter the repository name.

Select Add a README file.

Create the repository.

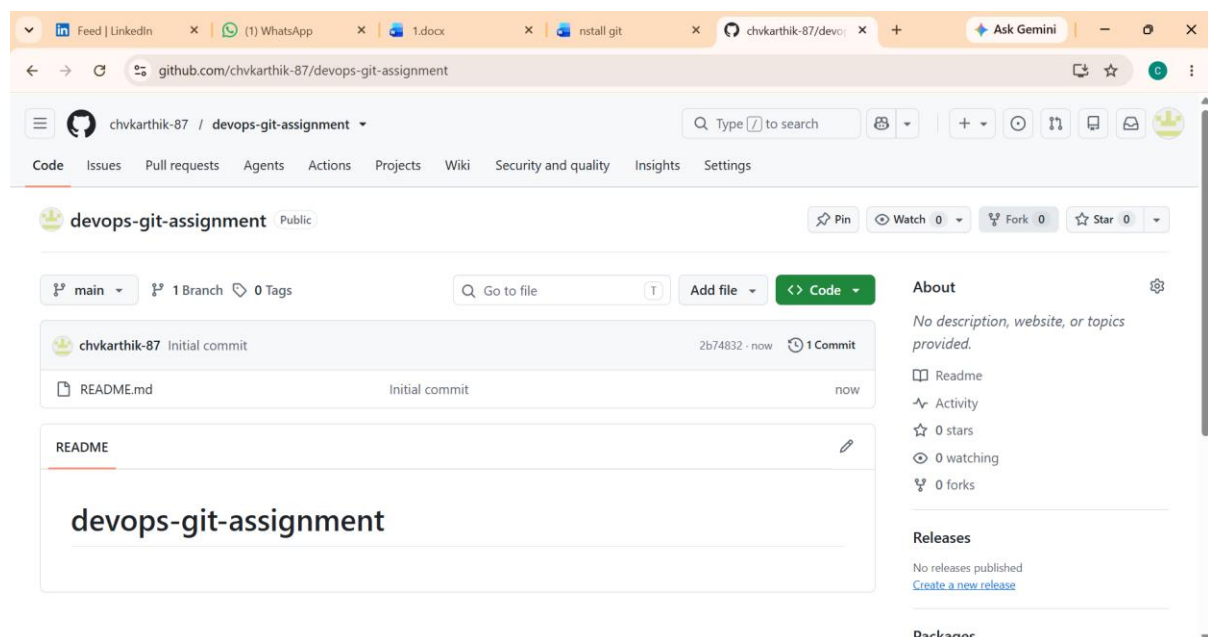
Create .gitignore if it was not created during repository creation.

Files

README.md

.gitignore

Description : README.md provides information about the repository, while .gitignore specifies files that Git should not track.



3. Clone the created repo to local.

Purpose :To create a local copy of the GitHub repository for performing Git operations.

Command : git clone <repository-url>

Example: git clone https://github.com/chvkarthik-87/devops-git-assignment.git

Then: cd devops-git-assignment

Description :The git clone command downloads the repository from GitHub to the local system, including its Git history.

```
chvka@karthik MINGW64 ~/git-task
$ git clone https://github.com/chvkarthik-87/devops-git-assignment.git
fatal: destination path 'devops-git-assignment' already exists and is not an empty directory.

chvka@karthik MINGW64 ~/git-task
$ cd devops-git-assignment/

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (main)
$ ls -la
total 4
drwxr-xr-x 1 chvka 197609 0 Aug 20 14:26 ./
drwxr-xr-x 1 chvka 197609 0 Aug 20 14:26 ../
drwxr-xr-x 1 chvka 197609 0 Aug 20 14:26 .git/
```

4. Create two files in local repo.

Purpose :To create new files inside the local Git repository.

Commands:

touch file1.txt file2.txt

Check the files:

ls

Description: the touch command creates empty files. These files are initially untracked by Git until they are added to the staging area.

Files Created: file1.txt file2.txt

```
chvka@karthik MINGW64 ~/git-task/devops-git-assignment (main)
$ touch file1.txt file2.txt

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (main)
$ ls
README.md  file1.txt  file2.txt

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (main)
$ |
```

5. Commit two files and push to central Repository.

Purpose :To save the changes in Git history and upload them to the GitHub repository.

Commands

git add .

git commit -m "Added two files"

git push origin main

Description

git add moves the files to the staging area.

git commit saves the changes in the local Git repository.

git push uploads the committed changes to GitHub.

```
chvka@karthik MINGW64 ~/git-task/devops-git-assignment (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

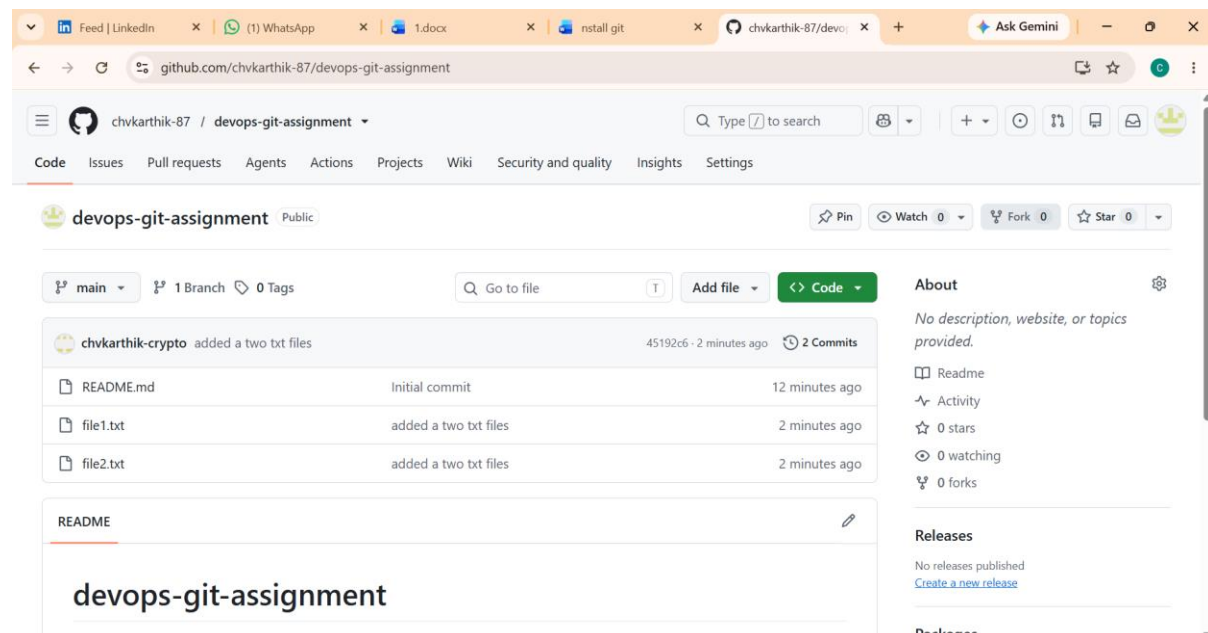
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file1.txt
        file2.txt

nothing added to commit but untracked files present (use "git add" to track)

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (main)
$ git add .

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (main)
$ git commit -m "added a two txt files"
[main 45192c6] added a two txt files
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 file1.txt
 create mode 100644 file2.txt

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (main)
$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 286 bytes | 143.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/chvkarthik-87/devops-git-assignment.git
 2b74832..45192c6  main -> main
```



6. Create a branch in local and create a sample file and push to central.

Commands

git branch release

git checkout release

Create a sample file:

touch sample.txt

Add and commit:

git add .

git commit -m "Added sample file"

Push:

git push -u release

Description

A branch allows developers to work on changes independently without directly modifying the main branch.

```
chvka@karthik MINGW64 ~/git-task/devops-git-assignment (main)
$ git branch release

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (main)
$ git checkout release
Switched to branch 'release'

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ touch sample.txt

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git add .

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git commit -m "added a sample file to release"
[release 3a50b4f] added a sample file to release
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 sample.txt

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git push origin release
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 245 bytes | 245.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'release' on GitHub by visiting:
remote:   https://github.com/chvkarthik-87/devops-git-assignment/pull/new/release
remote:
To https://github.com/chvkarthik-87/devops-git-assignment.git
 * [new branch]   release -> release

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git branch -a
  main
* release
remotes/origin/main
remotes/origin/release
```

7. Create a branch in github and clone that to local.

Purpose : To create a branch directly in GitHub and make that branch available in the local repository.

GitHub Steps

Open the GitHub repository.

Select the main branch dropdown.

Create a new branch named:

github-branch

Local Commands

git fetch origin

Create the local branch:

git branch github-branch origin/github-branch

Switch to it:

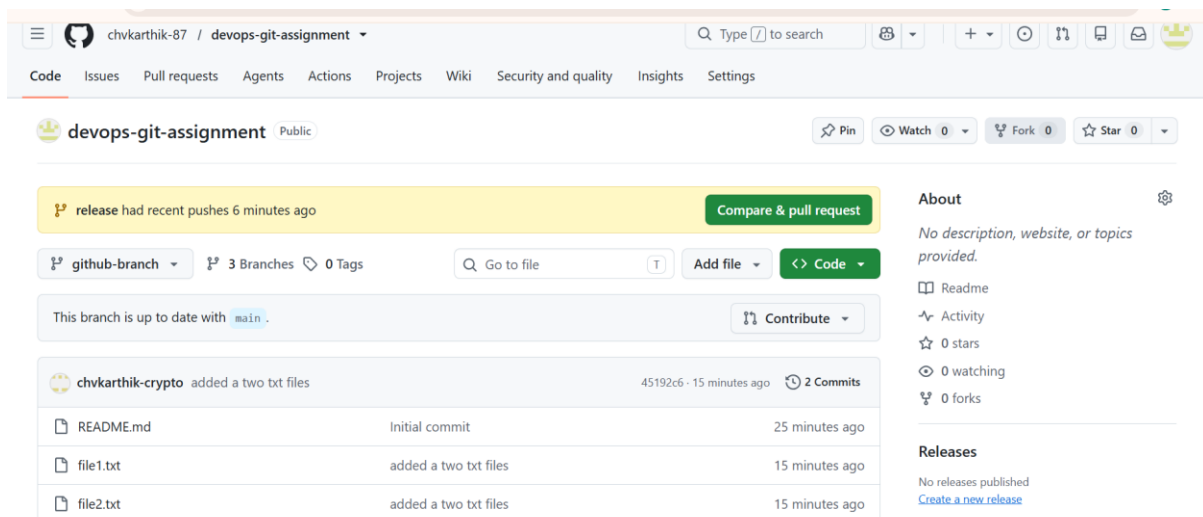
git checkout github-branch

Check:

git branch

Description

git fetch downloads information about the latest remote branches without modifying the working files.



```
chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git fetch origin
From https://github.com/chvkarthik-87/devops-git-assignment
* [new branch]      github-branch -> origin/github-branch

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git branch github-branch origin/github-branch
branch 'github-branch' set up to track 'origin/github-branch'.

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git checkout github-branch
Switched to branch 'github-branch'
Your branch is up to date with 'origin/github-branch'.

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (github-branch)
$ git branch
* github-branch
  main
  release

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (github-branch)
$
```

8. Merge the created branch with master in git local.

Purpose : To combine the changes from a feature branch into the main branch.

Commands

Switch to main:

git checkout main

Merge:

git merge release

Push the merged changes:

git push origin main

Description: The git merge command combines the history and changes of one branch into the currently checked-out branch.

```
chvka@karthik MINGW64 ~/git-task/devops-git-assignment (github-branch)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (main)
$ git merge release
Updating 45192c6..3a50b4f
Fast-Forward
 sample.txt | 0
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 sample.txt

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (main)
$ git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/chvkarthik-87/devops-git-assignment.git
 45192c6..3a50b4f  main -> main

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (main)
$ git log
commit 3a50b4f4127927225efff154aa3efffa82ec3c736 (HEAD -> main, origin/release, origin/main, release)
Author: karthik <chvkarthik07@gmail.com>
Date: Thu Aug 20 14:48:12 2026 +0530

    added a sample file to release

commit 45192c623212b88d0739921bf2ba09884008b4e7 (origin/github-branch, github-branch)
Author: karthik <chvkarthik07@gmail.com>
Date: Thu Aug 20 14:41:59 2026 +0530

    added a two txt files

commit 2b7483285d0692d22e8ee7ade4bd095a881abd78
Author: chvkarthik-87 <chvkarthik8989@gmail.com>
Date: Thu Aug 20 14:31:52 2026 +0530

    Initial commit
```

9. Merge the created branch with master in github by sending a pull request.

Purpose : To understand how changes are reviewed and merged into the main branch using a GitHub Pull Request.

Steps

Push a new change to feature-branch.

Open the GitHub repository.

Click Compare & pull request.

Select:

Base: main

Compare: feature-branch

Click Create pull request.

Click Merge pull request.

Click Confirm merge.

Description

A Pull Request allows changes from one branch to be reviewed before they are merged into the main branch.

```

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (main)
$ git checkout release
Switched to branch 'release'

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ touch pr-file.txt

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ ls
README.md file1.txt file2.txt pr-file.txt sample.txt

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git add .

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git commit -m "added a new pr-file to the release"
[release fac54f2] added a new pr-file to the release
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 pr-file.txt

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git push origin release
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 245 bytes | 245.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/chvkarthik-87/devops-git-assignment.git
3a50b4f..fac54f2 release -> release

```

github.com/chvkarthik-87/devops-git-assignment/compare/release?expand=1

base: main ← compare: release ✓ Able to merge. These branches can be automatically merged.

Add a title *

added a new pr-file to the release

Add a description

Write Preview

H B I

Add your description here...

☐ Markdown is supported
 ☐ Paste, drop, or click to add files

Create pull request

Reviewers
No reviews

Assignees
No one—[assign yourself](#)

Labels
None yet

Projects
None yet

Milestone
No milestone

Development
Use [Closing keywords](#) in the description to automatically close issues

Helpful resources

added a new pr-file to the release #1

chvkarthik-87 wants to merge 1 commit into `main` from `release`

Conversation 0 Commits 1 Checks 0 Files changed 1

chvkarthik-87 commented 2 minutes ago

No description provided.

added a new pr-file to the release

No conflicts with base branch
Merging can be performed automatically.

Merge pull request You can also merge this with the command line. [View command line instructions.](#)

Still in progress? Convert to draft

Reviewers
No reviews
Still in progress? [Convert to draft](#)

Assignees
No one—assign yourself

Labels
None yet

Projects
None yet

Milestone
No milestone

added a new pr-file to the release #1

chvkarthik-87 wants to merge 1 commit into `main` from `release`

Commit message

Merge pull request #1 from chvkarthik-87/release

Extended description

added a new pr-file to the release

This commit will be authored by chvkarthik8989@gmail.com.

Confirm merge Cancel

Still in progress? Convert to draft

Labels
None yet

Projects
None yet

Milestone
No milestone

Development
Successfully merging this pull request may close these issues.

None yet

Notifications
Customize
Unsubscribe
You're receiving notifications because you authored the thread.

Merged added a new pr-file to the release #1

chvkarthik-87 merged 1 commit into `main` from `release`

Pull request successfully merged and closed
You're all set — the `release` branch can be safely deleted.

Delete branch

Add a comment

Write Preview

Add your comment here...

Markdown is supported Paste, drop, or click to add files

Comment

Remember, contributions to this repository should follow our [GitHub Community Guidelines](#).

ProTip! Add comments to specific lines under [Files changed](#).

Projects
None yet

Milestone
No milestone

Development
Successfully merging this pull request may close these issues.

None yet

Notifications
Customize
Unsubscribe
You're receiving notifications because you modified the open/close state.

2 participants

Lock conversation

10. Create a file in local and send that to branch in github.

Purpose :To create a new file locally and upload it to a specific remote branch.

Commands

Switch to the branch:

git checkout release

Create a file:

touch newfile.txt

Add:

git add .

Commit:

git commit -m "Added new file"

Push:

git push origin release

Description :The file is created locally, committed to the local Git repository, and then pushed to the specified GitHub branch.

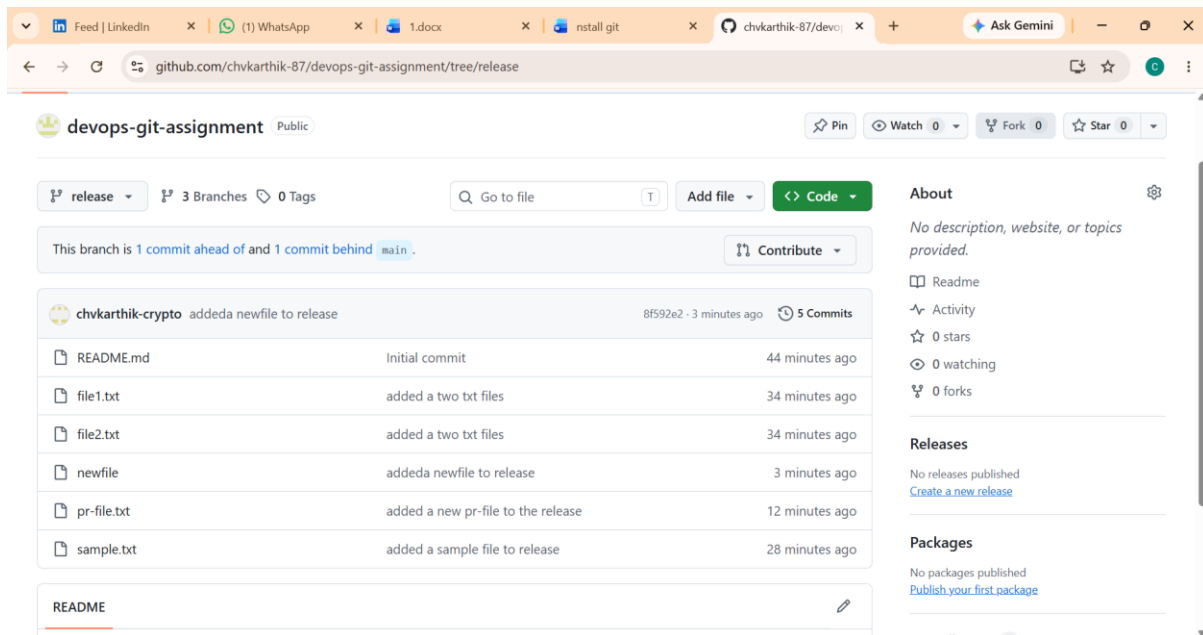
```
chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ touch newfile

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git add .

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git commit -m "addeda newfile to release"
[release 8f592e2] addeda newfile to release
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 newfile

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git push origin release
bash: git: command not found

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git push origin release
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 240 bytes | 240.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/chvkarthik-87/devops-git-assignment.git
fac54f2..8f592e2  release -> release
```



11. Clone only a branch from github to local.

Purpose : To work with a specific GitHub branch instead of the default branch.

Command

```
git clone https://github.com/chvkarthik-87/devops-git-assignment.git
```

Then:

```
cd devops-git-assignment
```

Fetch branches:

```
git fetch origin
```

Switch to the required branch:

```
git checkout release
```

Check:

```
git branch
```

Description : The repository is cloned locally and then the required branch is checked out for development

```
chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ cd ..

chvka@karthik MINGW64 ~/git-task
$ git clone https://github.com/chvkarthik-87/devops-git-assignment.git
fatal: destination path 'devops-git-assignment' already exists and is not an empty directory.

chvka@karthik MINGW64 ~/git-task
$ cd devops-git-assignment/

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git fetch origin
remote: Enumerating objects: 1, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (1/1), 903 bytes | 225.00 KiB/s, done.
From https://github.com/chvkarthik-87/devops-git-assignment
   3a50b4f..14ed980  main       -> origin/main

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git branch
  github-branch
* main
* release
```

12. Create a file with all passwords and make that untrackable with git.

Purpose : To create a password file and configure Git to ignore it using .gitignore, preventing the file from being tracked and pushed to GitHub.

Steps

1. Create the password file : touch passwords.txt

2. Open the .gitignore file : nano .gitignore

Add: passwords.txt

Save and exit:

Ctrl + O

Enter

Ctrl + X

3. Verify the .gitignore file : cat .gitignore

Expected output: passwords.txt

4. Check Git status : git status

The passwords.txt file should not appear as an untracked file.

5. Commit the .gitignore file

git add .gitignore

git commit -m "Added password file to gitignore"

6. Verify the repository status:

git status

```

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ touch password.txt

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ ls
README.md  file1.txt  file2.txt  newfile  password.txt  pr-file.txt  sample.txt

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ nano .gitignore

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ cat .gitignore
password.txt

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git status
On branch release
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore

nothing added to commit but untracked files present (use "git add" to track)

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git add .
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git commit -m "Added password file to gitignore"
[release 25f3892] Added password file to gitignore
1 file changed, 1 insertion(+)
create mode 100644 .gitignore

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git status
On branch release
nothing to commit, working tree clean

```

13. Make a commit and make that commit reset without savings changes.

Purpose : To understand how to remove the latest commit and permanently discard its changes using git reset --hard.

Step 1 – Create a test file

touch resetfile.txt

Step 2 – Add the file

git add resetfile.txt

Step 3 – Commit the file

git commit -m "Added reset file"

Step 4 – Check the commit

git log --oneline

You should see your latest commit at the top.

Step 5 – Reset the commit and discard changes

git reset --hard HEAD~1

Then:

git status

```

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ touch resetfile.txt

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git add .

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git commit -m "Added reset file to release"
[release 1675d23] Added reset file to release
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 resetfile.txt

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git log
commit 1675d2330d16cae5ed30f8f6cdb79f6f2864afc7 (HEAD -> release)
Author: karthik <chvkarthik07@gmail.com>
Date: Thu Aug 20 15:59:45 2026 +0530

    Added reset file to release

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git reset --hard HEAD~1
HEAD is now at 25f3892 Added password file to gitignore

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git status
On branch release
nothing to commit, working tree clean

```

14. Revert a committed commit to the older version.

Purpose : To undo the changes introduced by a previous commit while keeping the Git history intact.

Step 1 – Check the commit history

git log --oneline

You will see commit IDs like:

abc1234 Updated file

def5678 Added file

Copy the commit ID of the commit you want to undo.

Step 2 – Revert the commit

git revert abc1234

Replace abc1234 with your actual commit ID.

If an editor opens, save and exit:

Ctrl + O

Enter

Ctrl + X

Step 3 – Check the log

git log --oneline

You will see a new commit similar to:

Revert "Updated file"

Step 4 – Check the status

git status

Step 5 – Push the revert to GitHub

If the original commit was already pushed to GitHub:

git push origin main

Result :The selected commit's changes were successfully undone by creating a new revert commit. The original commit remains in the Git history.

```
chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git log --oneline
8f592e2 (HEAD -> release, origin/release) added a new file to release
fac54f2 added a new pr-file to the release
3a50b4f (main) added a sample file to release
45192c6 (origin/github-branch, github-branch) added a two txt files
2b74832 Initial commit

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git revert 3a50b4f
[release a49ca57] Revert "added a sample file to release"
1 file changed, 0 insertions(+), 0 deletions(-)
delete mode 100644 sample.txt

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git log --oneline
a49ca57 (HEAD -> release) Revert "added a sample file to release"
8f592e2 (origin/release) added a new file to release
fac54f2 added a new pr-file to the release
3a50b4f (main) added a sample file to release
45192c6 (origin/github-branch, github-branch) added a two txt files
2b74832 Initial commit

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git status
On branch release
Untracked files:
  (use "git add <file>..." to include in what will be committed)
      password.txt

nothing added to commit but untracked files present (use "git add" to track)

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git push origin release
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 278 bytes | 278.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/chvkarthik-87/devops-git-assignment.git
   8f592e2..a49ca57  release -> release
```

15. Push a file to stash without saving the changes and work on another file.

Purpose :To temporarily save unfinished changes using Git Stash, work on another file, and continue the previous work later.

Step 1 — Modify a file

Use an existing file:

nano file1.txt

Add any simple text, for example:

Stash practice

Save and exit:

Ctrl + O → Enter → Ctrl + X

Step 2 — Check the changes

git status

file1.txt should show as modified.

Step 3 — Stash the changes

git stash

Now the changes in file1.txt are temporarily stored and removed from the working directory.

Check:

git status

Step 5 — Check the stash

git stash list

You should see something like:

stash@{0}: WIP on release: ...

Result :The changes made to file1.txt were temporarily stored using git stash, allowing work to continue on file2.txt without losing the unfinished changes.

```
chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ nano file1.txt

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git status
On branch release
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   file1.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        password.txt

no changes added to commit (use "git add" and/or "git commit -a")

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git stash
warning: in the working copy of 'file1.txt', LF will be replaced by CRLF the next time Git touches it
Saved working directory and index state WIP on release: a49ca57 Revert "added a sample file to release"

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git status
On branch release
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        password.txt

nothing added to commit but untracked files present (use "git add" to track)

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git stash list
stash@{0}: WIP on release: a49ca57 Revert "added a sample file to release"
```

16. Undo the stash file and start working on that again.

Purpose : To restore the previously stashed changes and continue working on the file.

Step 1 — Check the stash

git stash list

You should see:

stash@{0}.....

Step 2 — Restore the stashed changes

git stash pop

This brings back the changes that you made to file1.txt.

Step 3 — Check the status

git status

You should see:

modified: file1.txt

Step 4 — Continue working on the file

nano file1.txt

Add another line, for example:

Continuing stash practice

Save and exit:

Ctrl + O → Enter → Ctrl + X

Step 5 — Check the changes

git diff

Result :The previously stashed changes were successfully restored using git stash pop, and the work on file1.txt was continue

```
chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git stash list
stash@{0}: WIP on release: a49ca57 Revert "added a sample file to release"

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git stash pop
On branch release
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   file1.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        password.txt

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (aac733707ccbb3630140bdfbe21a48d028d17e89)

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git status
On branch release
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   file1.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        password.txt

no changes added to commit (use "git add" and/or "git commit -a")

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ nano file1.txt

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git diff
diff --git a/file1.txt b/file1.txt
index e69de29..83cf59e 100644
--- a/file1.txt
+++ b/file1.txt
@@ -0,0 +1 @@
+continue stash practice
```

17. Generate a ssh-keygen and configure into github.

Purpose :To generate an SSH key and configure it with GitHub for secure authentication.

Steps

1. Generate the SSH key

Run:

ssh-keygen

When the following prompts appear, press Enter for each:

Enter file in which to save the key:

Enter passphrase:

Enter same passphrase again:

2. Check the generated SSH key

ls ~/.ssh

Expected output:

id_rsa

id_rsa.pub

3. Display the public key

```
cat ~/.ssh/id_rsa.pub
```

Copy the complete output.

4. Add the SSH key to GitHub

Go to:

GitHub → Settings → SSH and GPG keys → New SSH key

Enter a title, paste the copied public key, and click Add SSH key.

5. Test the SSH connection

Run:

```
ssh -T git@github.com
```

If GitHub asks:

Are you sure you want to continue connecting?

Type:

yes

Expected Result

A successful authentication message similar to:

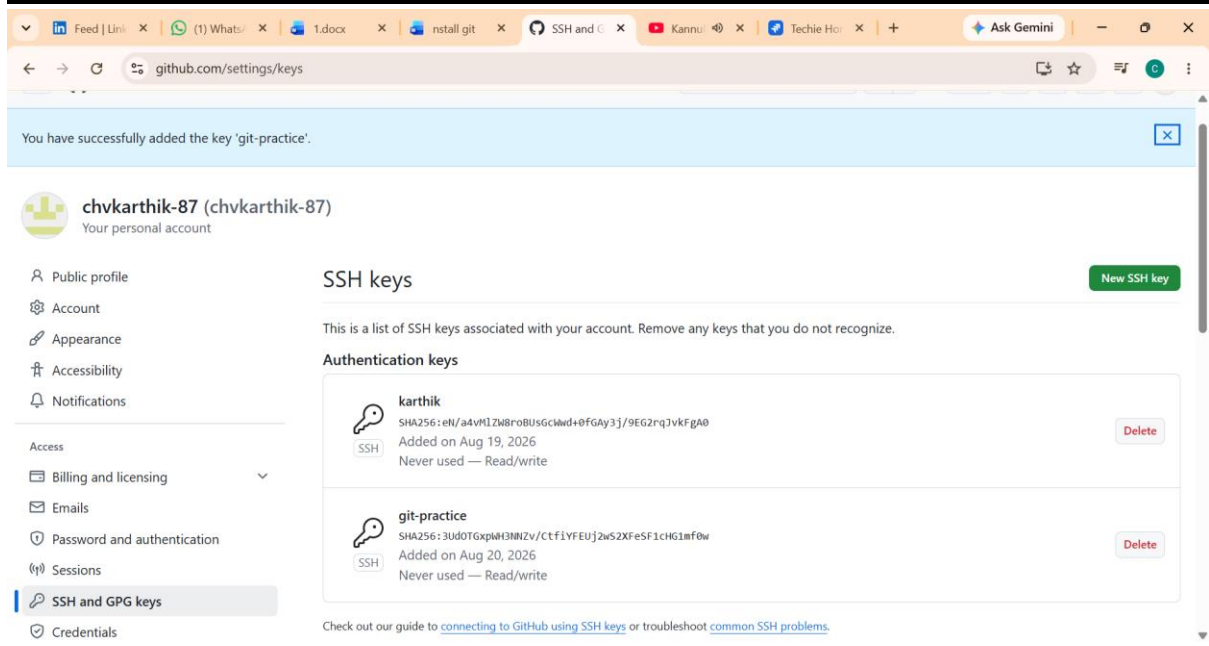
Hi chvkarthik-87! You've successfully authenticated, but GitHub does not provide shell access.

```
chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/c/Users/chvka/.ssh/id_rsa):
/c/Users/chvka/.ssh/id_rsa already exists.
Overwrite (y/n)? yes
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/chvka/.ssh/id_rsa
Your public key has been saved in /c/Users/chvka/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:3Ud0TGxpWH3NNZv/CtfiYFEUj2wS2XFeSF1cHG1mf0w chvka@karthik
The key's randomart image is:
+---[RSA 3072]-----+
|      .BB&&|
|      o*B=E|
|      .oOO=|
|      . . B =|
|    S . o o o|
|      o . .|
|      + o o|
|      . = o|
|      o    |
+-----[SHA256]-----+
```

```
chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ ls ~/.ssh
id_rsa id_rsa.pub known_hosts known_hosts.old

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQgCOrNLFM3qGg9prKujbA8xNrdCU8GRxnHUGRQb5F6Tcz0+hFoDqZnORbGNGoFBtctTAGbreDfGB4qnm5mkcLZ66AqEj29jc/tjn1guUB9nG9CBibGqf2IFUuwHwSF/k4xTDpQOB0F1KD3+HB82RK3jnnto7rwcGgPs9Gs2k5FKntqp04GMNJ09cdm0x1S1dhy7a9eCg5evpgwa4V1TuMqRTuvvtzJ2EaYAvFmPK1aLMP/Sy8At2YRFcthoUrdOM2HuCTE/+ho31XTBcdWdy4BzwwcEmkzr6bKKZB02oz/1MOZNG8V94uPrt/D5okE1vx1hdd1yDPPaR5YQDN8xmg/ej1qgCrTR0MH0AqCjoqIGotZpwgDh2UDG1anueR5Ug+d08JmgjFncG5Bm3IAZQnqXBOJrxXk4PD3TnnRpQFEE1o0GEBFuJ10A+3X1FTRbt7EjksRVQlw05zM4VkuXRRSP4TVYeRQ/owDon5yAB5qw6kg5ynjMhGVnbaNfaJh18= chvka@karthik

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ ssh -T git@github.com
The authenticity of host 'github.com (20.207.73.82)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvV6TuJJhpbZisF/zLDA0zPMSvHdkr4UvCOQU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Hi chvkarthik-87! You've successfully authenticated, but GitHub does not provide shell access.
```



The screenshot shows the GitHub 'SSH keys' page for user 'chvkarthik-87'. A success message at the top states 'You have successfully added the key 'git-practice''. The page lists two SSH keys: 'karthik' (added Aug 19, 2026) and 'git-practice' (added Aug 20, 2026). Both keys are marked as 'Never used' and have 'Read/write' permissions. A 'New SSH key' button is visible in the top right. The left sidebar shows the user's profile and navigation links, with 'SSH and GPG keys' selected.

```
chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ ssh -T git@github.com
The authenticity of host 'github.com (20.207.73.82)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvV6TuJJhpbZisF/zLDA0zPMSvHdkr4UvCOQU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Hi chvkarthik-87! You've successfully authenticated, but GitHub does not provide shell access.
```

18. Configure webhooks to github.

Purpose :To configure a webhook in the GitHub repository and send repository events to an external webhook URL.

Steps

1. Open the GitHub repository

Open the devops-git-assignment repository.

2. Open Webhooks

Go to:

Settings → Webhooks → Add webhook

3. Enter the Payload URL

A webhook URL was generated using Webhook.site.

Example:

<https://webhook.site/xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxxxx>

Enter the generated URL in the Payload URL field.

4. Select Content Type

Select:

application/json

5. Secret

Leave the Secret field blank.

6. Select the Event

Select:

Just the push event

7. SSL Verification

Keep:

Enable SSL verification
enabled.

8. Activate the Webhook

Keep the Active option enabled.

9. Add the Webhook

Click:

Add webhook

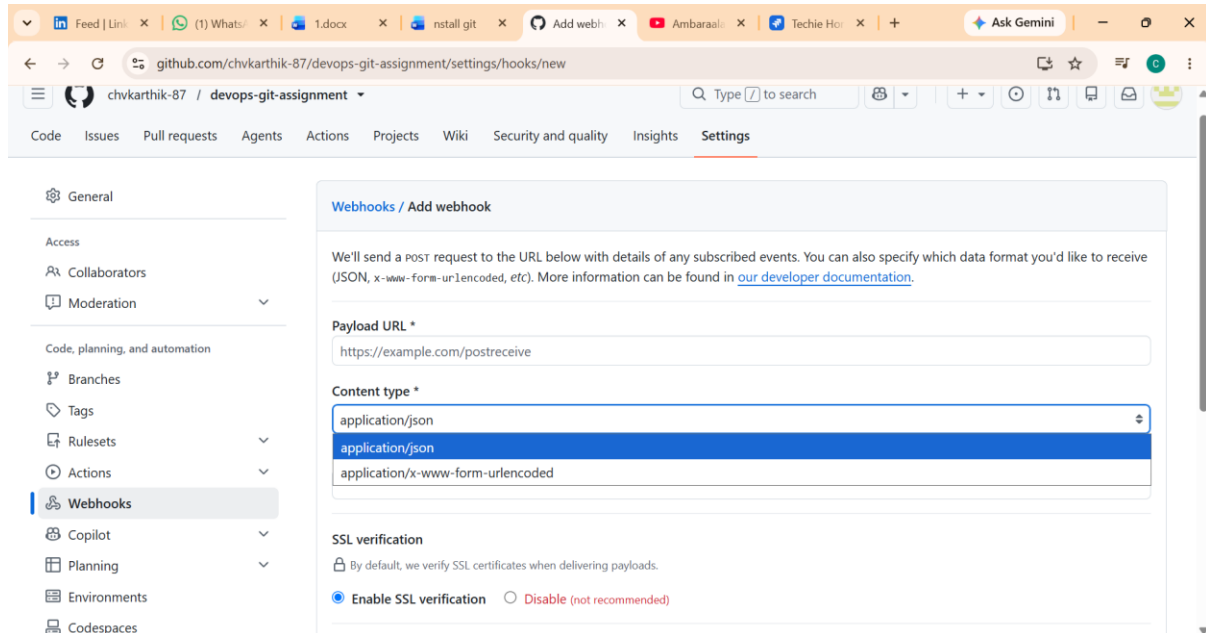
Verification

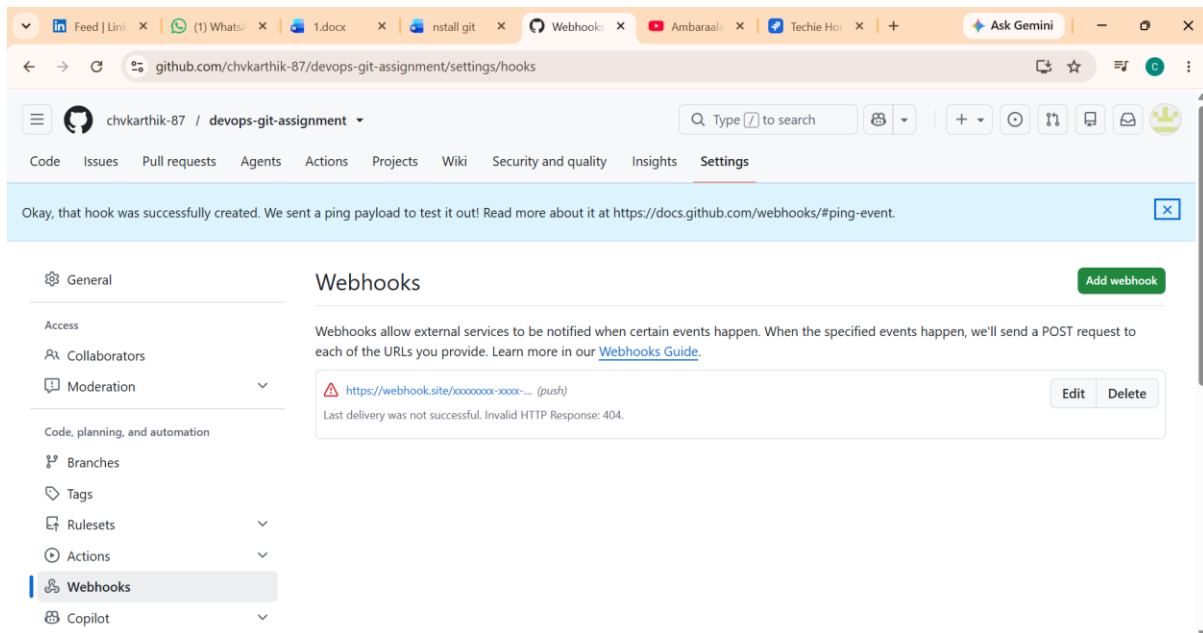
After adding the webhook, make a small push to the GitHub repository.

The webhook testing page should receive a request from GitHub.

Result

The GitHub webhook was successfully configured to send push event notifications to the specified webhook URL.





19. Basic understanding of .git file.

Purpose : To understand the .git directory and its role in a Git repository.

Step 1 – Check the .git directory

Inside your repository, run:

```
ls -la
```

You should see:

```
.git
```

Step 2 – View the .git directory

```
ls -la .git
```

You will see files and folders such as:

```
HEAD
```

```
config
```

```
objects
```

```
refs
```

```
index
```

```
logs
```

Description

The .git directory contains the internal information required by Git to manage the repository.

HEAD → Points to the current branch.

config → Contains repository-specific Git configuration.

objects → Stores Git objects such as commits and file data.

refs → Stores references to branches and tags.

index → Stores staging area information.

logs → Contains information about changes to references.

⚠ Do not delete or manually modify the .git directory, because it can damage the Git repository.

Result :The .git directory and its basic structure were successfully examined and understood.

```
chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ ls -la
total 10
drwxr-xr-x 1 chvka 197609  0 Aug 20 16:27 ./
drwxr-xr-x 1 chvka 197609  0 Aug 20 14:26 ../
drwxr-xr-x 1 chvka 197609  0 Aug 20 16:28 .git/
-rw-r--r-- 1 chvka 197609 23 Aug 20 14:39 README.md
-rw-r--r-- 1 chvka 197609 25 Aug 20 16:28 file1.txt
-rw-r--r-- 1 chvka 197609  0 Aug 20 14:40 file2.txt
-rw-r--r-- 1 chvka 197609  0 Aug 20 15:13 newfile
-rw-r--r-- 1 chvka 197609  0 Aug 20 15:50 password.txt
-rw-r--r-- 1 chvka 197609  0 Aug 20 15:03 pr-file.txt

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ ls -la .git
total 24
drwxr-xr-x 1 chvka 197609  0 Aug 20 16:28 ./
drwxr-xr-x 1 chvka 197609  0 Aug 20 16:27 ../
-rw-r--r-- 1 chvka 197609  41 Aug 20 16:27 AUTO_MERGE
-rw-r--r-- 1 chvka 197609 351 Aug 20 16:10 COMMIT_EDITMSG
-rw-r--r-- 1 chvka 197609 393 Aug 20 15:19 FETCH_HEAD
-rw-r--r-- 1 chvka 197609  24 Aug 20 15:12 HEAD
-rw-r--r-- 1 chvka 197609  41 Aug 20 16:23 ORIG_HEAD
-rw-r--r-- 1 chvka 197609 393 Aug 20 14:56 config
-rw-r--r-- 1 chvka 197609  73 Aug 20 14:26 description
drwxr-xr-x 1 chvka 197609  0 Aug 20 14:26 hooks/
-rw-r--r-- 1 chvka 197609 414 Aug 20 16:27 index
drwxr-xr-x 1 chvka 197609  0 Aug 20 14:26 info/
drwxr-xr-x 1 chvka 197609  0 Aug 20 14:39 logs/
drwxr-xr-x 1 chvka 197609  0 Aug 20 16:23 objects/
drwxr-xr-x 1 chvka 197609  0 Aug 20 16:27 refs/
```

20. Check all the logs of git.

Purpose :To view the complete commit history of the Git repository.

Step 1 – Check Git logs

Run:

git log

This displays details such as:

Commit ID

Author

Date

Commit message

Result :The Git commit history was successfully viewed, including commits from different branches.

```

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git log
commit a49ca57dffeabaf953cab6104be7eaea112d4b3cf (HEAD -> release, origin/release)
Author: karthik <chvkarthik07@gmail.com>
Date: Thu Aug 20 16:08:26 2026 +0530

    Revert "added a sample file to release"

    This reverts commit 3a50b4f4127927225efff154aa3effa82ec3c736.

commit 8f592e271dc5ebea974e4f1625db7113b6f155c8
Author: karthik <chvkarthik07@gmail.com>
Date: Thu Aug 20 15:13:37 2026 +0530

    added a newfile to release

commit fac54f28e672cc5350461d18f4ed1a5d90d2ee7f
Author: karthik <chvkarthik07@gmail.com>
Date: Thu Aug 20 15:04:46 2026 +0530

    added a new pr-file to the release

commit 3a50b4f4127927225efff154aa3effa82ec3c736 (main)
Author: karthik <chvkarthik07@gmail.com>
Date: Thu Aug 20 14:48:12 2026 +0530

    added a sample file to release

commit 45192c623212b88d0739921bf2ba09884008b4e7 (origin/github-branch, github-branch)
Author: karthik <chvkarthik07@gmail.com>
Date: Thu Aug 20 14:41:59 2026 +0530

    added a two txt files

commit 2b7483285d0692d22e8ee7ade4bd095a881abd78
Author: chvkarthik-87 <chvkarthik8989@gmail.com>
Date: Thu Aug 20 14:31:52 2026 +0530

    Initial commit

```

21. Rename the commit message.

Purpose :To change the message of the most recent Git commit.

Step 1 – Check the latest commit

git log --oneline

Example:

abc1234 Added new file

Step 2 – Rename the commit message

git commit --amend -m "Updated commit message"

Step 3 – Verify

git log --oneline

Now the latest commit message should be:

abc1234 Updated commit message

Result :The latest Git commit message was successfully renamed using git commit --amend.

```

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git log --oneline
a49ca57 (HEAD -> release, origin/release) Revert "added a sample file to release"
8f592e2 added a new file to release
fac54f2 added a new pr-file to the release
3a50b4f (main) added a sample file to release
45192c6 (origin/github-branch, github-branch) added a two txt files
2b74832 Initial commit

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git commit --amend -m "updated commit message"
[release 4a0cbc3] updated commit message
Date: Thu Aug 20 16:08:26 2026 +0530
1 file changed, 0 insertions(+), 0 deletions(-)
delete mode 100644 sample.txt

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git log --oneline
4a0cbc3 (HEAD -> release) updated commit message
8f592e2 added a new file to release
fac54f2 added a new pr-file to the release
3a50b4f (main) added a sample file to release
45192c6 (origin/github-branch, github-branch) added a two txt files
2b74832 Initial commit

```

22. Merge multiple commits into single commit.

Purpose : To combine multiple Git commits into a single commit using interactive rebase.

Steps

1. Check the Git status

git status

If there are uncommitted changes, temporarily save them using:

git stash

2. Start the interactive rebase

git rebase -i HEAD~3

The interactive rebase editor opens with the recent commits.

Keep the first commit as:

pick

Change the other commits to:

squash

Example:

pick commit1

squash commit2

squash commit3

Save and exit:

Ctrl + O → Enter → Ctrl + X

3. Enter the final commit message

When Git asks for the final commit message, enter:

Combined multiple commits

Save and exit:

Ctrl + O → Enter → Ctrl + X

4. Verify the rebase

git log --oneline -5

The selected commits should now be combined into a single commit.

5. Restore the previously stashed changes**If git stash was used:**

git stash pop

Then check:

git status

Result :Multiple commits were successfully combined into a single commit using Git interactive rebase.

```

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git status
On branch release
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   file1.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        password.txt

no changes added to commit (use "git add" and/or "git commit -a")

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git stash
Saved working directory and index state WIP on release: 4a0cbc3 updated commit message

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git rebase -i HEAD~3
[detached HEAD e43894f] added a new pr-file to the release
Date: Thu Aug 20 15:04:46 2026 +0530
2 files changed, 0 insertions(+), 0 deletions(-)
rename sample.txt => newfile (100%)
create mode 100644 pr-file.txt
Successfully rebased and updated refs/heads/release.

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git log --oneline -5
e43894f (HEAD -> release) added a new pr-file to the release
3a50b4f (main) added a sample file to release
45192c6 (origin/github-branch, github-branch) added a two txt files
2b74832 Initial commit

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git stash pop
On branch release
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   file1.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        password.txt

no changes added to commit (use "git add" and/or "git commit -a")

```

```

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (87054c633bd35edee444bb4b07fa46e9b98ae247)

chvka@karthik MINGW64 ~/git-task/devops-git-assignment (release)
$ git status
On branch release
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   file1.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        password.txt

no changes added to commit (use "git add" and/or "git commit -a")

```